

Ridge Regression: From First Principles

Why Do We Need Ridge Regression?

Linear Regression tries to find the best line or hyperplane by choosing coefficients that make prediction errors small. This works well when the relationship is simple and the features are not causing too much instability. But in real datasets, Linear Regression can become too confident. It may give very large coefficient values to some features, especially when features are correlated or when the model is trying too hard to fit training data.

Imagine a student preparing for an exam. A good student understands the concepts. An overfitting student memorizes every example from the practice sheet, including unusual questions that may never appear again. Linear Regression can behave like that when it bends too much toward the training data. The model may perform well on training examples but poorly on new examples.

Ridge Regression was created to control this problem. It keeps the basic Linear Regression idea, but adds a penalty for large coefficients. The model is still allowed to learn from the data, but it is discouraged from making any coefficient unnecessarily large.

Overfitting and Large Coefficients

In Linear Regression, each coefficient tells us how strongly one feature affects the prediction. If a coefficient becomes very large, the prediction becomes very sensitive to that feature.

Suppose a house price model uses size, bedrooms, bathrooms, distance from city center, and many other features. If the coefficient for one feature becomes extremely large, a small change in that feature can create a huge change in predicted price. Sometimes this is real, but often it is a sign that the model is reacting too strongly to noise.

Ridge Regression reduces this sensitivity by shrinking coefficients toward zero. It usually does not make them exactly zero. It simply makes them smaller and more controlled.

The Core Idea of Ridge Regression

Ridge Regression uses the same prediction equation as Linear Regression.

$$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_n x_n$$

The difference is in the cost function. Linear Regression only tries to reduce prediction error. Ridge Regression tries to reduce prediction error and keep coefficients small.

The Ridge cost function is:

$$J(\theta) = MSE + \lambda \sum_{j=1}^n \theta_j^2$$

Here, MSE measures prediction error. The term $\sum_{j=1}^n \theta_j^2$ adds the squares of all feature coefficients. The value λ controls how strongly we punish large coefficients.

The intercept θ_0 is usually not penalized because it only shifts the prediction baseline. Ridge mainly controls the feature coefficients.

Understanding the L2 Penalty

Ridge Regression uses an L2 penalty. L2 means the model squares the coefficients before adding them.

$$L2 \text{ Penalty} = \theta_1^2 + \theta_2^2 + \dots + \theta_n^2$$

Squaring has an important effect. Large coefficients receive a much stronger penalty than small coefficients. For example, a coefficient of 2 contributes 4 to the penalty, but a coefficient of 10 contributes 100.

This pushes the model toward smaller, smoother coefficients. Ridge does not usually remove features completely. Instead, it spreads the effect across features more gently.

Lambda

The symbol λ controls the strength of regularization.

If $\lambda = 0$, Ridge Regression becomes ordinary Linear Regression because there is no penalty.

If λ is small, the model is only slightly discouraged from using large coefficients.

If λ is large, the model strongly avoids large coefficients. This can reduce overfitting, but if λ becomes too large, the model may become too simple and underfit.

So λ controls the balance between fitting the training data and keeping the model simple.

Bias-Variance Tradeoff

Ridge Regression is a good example of the bias-variance tradeoff.

A very flexible model may have low bias but high variance. It can fit the training data closely, but it may change a lot when the data changes.

A very simple model may have high bias but low variance. It is stable, but it may miss important patterns.

Ridge adds a little bias by shrinking coefficients, but it often reduces variance much more. This can improve test performance.

The goal is not to make coefficients as small as possible. The goal is to find a useful balance where the model generalizes well.

Geometric Intuition

Linear Regression searches for coefficients that minimize error. Ridge Regression searches for coefficients that minimize error while staying inside a smooth constraint region.

The Ridge constraint has a circular or spherical shape because it is based on squared coefficients.

For two coefficients, the constraint looks like:

$$\theta_1^2 + \theta_2^2 \leq t$$

This smooth shape is why Ridge tends to shrink coefficients gradually rather than forcing them exactly to zero. The model keeps most features, but reduces their strength.

Worked Example

Suppose two models have similar training error.

Model	MSE	Coefficients	L2 Penalty
A	10	2, 3	$2^2 + 3^2 = 13$
B	9	20, 30	$20^2 + 30^2 = 1300$

Ordinary Linear Regression may prefer Model B because its MSE is slightly smaller. Ridge Regression may prefer Model A because Model B uses huge coefficients and is likely too sensitive.

If $\lambda = 0.1$, the Ridge costs become:

$$Cost_A = 10 + 0.1(13) = 11.3$$

$$Cost_B = 9 + 0.1(1300) = 139$$

Ridge chooses the model that balances accuracy and simplicity.

Hyperparameters

The most important hyperparameter is λ , often called α in machine learning libraries. A small value behaves more like Linear Regression. A large value creates stronger coefficient shrinkage.

Feature scaling is important for Ridge Regression. If features use different scales, the penalty becomes unfair. A feature measured in thousands may get penalized differently from a feature measured between 0 and 1. Scaling puts features on comparable ranges before regularization is applied.

Advantages, Disadvantages, and Evaluation

Ridge Regression is useful when Linear Regression overfits, when features are correlated, or when we want a more stable model. It often improves generalization without changing the basic linear structure.

Its main disadvantage is that it usually does not perform feature selection. It shrinks coefficients but keeps most of them. If the goal is to remove unimportant features completely, Lasso may be more suitable.

Ridge is still a regression model, so it is evaluated using regression metrics such as MAE, MSE, RMSE, and R^2 .

Use Ridge when the relationship is mostly linear but ordinary Linear Regression is unstable or overfitting. Avoid relying on Ridge when the true pattern is highly non-linear unless feature engineering or another model is used.